

SACHIN DEVARAJAN

Data Analyst | Machine Learning Engineer | GenAI Engineer

sachindevarajan7@gmail.com | 8220226139 | Pudukkottai, Tamil Nadu, India | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Data professional with hands-on internship experience applying Python, SQL, Power BI, Tableau, Excel, and Machine Learning to solve business problems. Skilled in end-to-end data analysis, EDA, statistical analysis, feature engineering, dashboard development, KPI reporting, and predictive modeling, with growing experience in Generative AI and Retrieval-Augmented Generation (RAG) pipelines. Built an NLP sentiment classifier at 90% accuracy and Power BI dashboards spanning multiple linked datasets that surfaced measurable revenue, profitability, and operations insights. Seeking to deliver data-driven impact as a Data Analyst, Machine Learning Engineer, AI/GenAI Engineer, or BI Analyst.

EXPERIENCE

Data Science Intern — PySpiders

Dec 2025 – Present

- Engineered and validated structured datasets using Python, Pandas, and NumPy, improving data quality and reliability for downstream analytics workflows.
- Performed Exploratory Data Analysis (EDA) on large datasets to identify trends, anomalies, missing values, and business opportunities, informing data-driven decisions.
- Developed and validated multiple machine learning models using Scikit-learn, comparing algorithm performance through cross-validation and hyperparameter tuning (GridSearchCV).
- Collaborated with mentors on real-world analytics projects, following industry-standard data workflows and best practices for reproducibility.

PROJECTS

AI-Powered PDF Question Answering System (RAG-based) | Python, FAISS, Sentence Transformers, PyMuPDF

<https://github.com/SachinDevarajan/Python-chatbot>

- Developed an AI-powered PDF Question Answering system using Retrieval-Augmented Generation (RAG) concepts to retrieve context-aware answers from PDF documents.
- Built an end-to-end NLP pipeline covering PDF text extraction, preprocessing, chunking, semantic embeddings, and similarity search for efficient document retrieval.
- Implemented a FAISS vector database with Sentence Transformers embeddings to perform fast Top-K semantic search over document content.

IMDB Movie Review Sentiment Analysis | Python, NLP, Scikit-learn | https://github.com/SachinDevarajan/IMDB_Review_Analysis

- Built a machine learning model to classify 50,000+ movie reviews into positive and negative sentiment using NLP techniques.
- Converted text into numerical features using TF-IDF Vectorization and trained a Logistic Regression classifier through a full ML pipeline, with model serialization via Joblib.
- Achieved 90% test accuracy with 0.90 Precision, 0.90 Recall, and a 0.90 F1-Score, validated through a full classification report.

Blinkit Executive Dashboard | Power BI, DAX | <https://github.com/SachinDevarajan/Blinkit>

- Built a centralized executive Power BI dashboard across 5 linked datasets, cleaning records and designing a Star Schema data model with an optimized Date Dimension for fast query performance.
- Created 15+ DAX measures including Total Revenue, Average Order Value, Delivery Success Rate, Customer Retention Rate, ROAS, and MoM/YoY Growth for stakeholder reporting.
- Uncovered that email campaigns achieved the highest ROAS (1.97) and delivery success rate reached 69.4%, directly informing marketing and operations decisions.

Amazon Sales Dashboard | Power BI | https://github.com/SachinDevarajan/amazon_sales

- Designed an interactive Power BI dashboard giving stakeholders a unified view of sales performance, profitability, and regional trends.
- Built Star Schema relationships and DAX measures (Total Sales, Total Profit, Profit Margin, Average Sales, YoY Growth, Monthly Sales Trend) with KPI cards and drill-through reports.
- Identified top-selling product categories, high-performing regions, and profit-driving customer segments, while flagging low-performing products for improvement.

TECHNICAL SKILLS

Programming: Python (OOP, Functions, File/Exception Handling, Regex), SQL

Data Analytics: Data Cleaning, Data Wrangling, EDA, Statistical Analysis, Root Cause Analysis, Data Storytelling, Pandas, NumPy

Machine Learning: Regression, Classification, Clustering, Random Forest, XGBoost, SVM, K-Means, Ensemble Learning, ML Pipelines, GridSearchCV, Model Evaluation

Generative AI: Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Semantic Search, Sentence Transformers, FAISS, Vector Databases, Vector Embeddings, Text Chunking

Visualization & BI: Power BI, DAX, Star Schema, Tableau, Excel, Advanced Excel, KPI Reporting, Dashboard Development, Business Reporting

Databases: SQL, Advanced SQL, Joins, Window Functions, CTE, Stored Procedures, Query Optimization, Snowflake

Tools: Git, GitHub, PySpark, Hadoop HDFS, Databricks, Jupyter Notebook

CERTIFICATIONS

- HackerRank SQL (Intermediate) — HackerRank, 2026
- Deloitte Data Analysis Job Simulation — Deloitte/Forage, 2026
- Tata Data Visualisation: Empowering Business with Effective Insights — Tata/Forage, 2026

EDUCATION

M.Sc. Information Technology — Rathinam College, Coimbatore (CGPA: 6.80)

2022 – 2024

B.Sc. Computer Science — J.J. College, Pudukkottai (CGPA: 7.44)

2019 – 2022